

In some datasets, however, it may be that $\mathbf{x}_i$ lies on a much lower, nonlinear manifold. In this situation, a linear mapping would require more latent dimensions to represent the underlying structure than a nonlinear method. To accommodate these nonlinearities, one option is to use a multi-layer feedforward neural network $\mu_\theta : \mathbb{R}^K \rightarrow \mathbb{R}^G$ for the mapping from the latent variables $\mathbf{z}_i$ to the observed data $\mathbf{x}_i$. Such a neural network with $L \in \mathbb{N}$ layers has the form:

$$\mu_\theta(\mathbf{z}) = W_{L+1}a_{b_L}(W_L a_{b_{L-1}} \cdots a_{b_1}(W_1 \mathbf{z})) + b_{L+1}, \quad (29)$$

where p_l is the dimension of layer l , $b_l \in \mathbb{R}^{p_l}$ are shift vectors, $W_l \in \mathbb{R}^{p_l \times p_{l-1}}$ are weight matrices and a_{b_l} is an activation function. The collection of latent variables is denoted by $\theta = \{b_l, W_l\}_{l=1}^{L+1}$.

We can then model the data using this flexible neural network mapping in the following deep generative model (DGM) (Kingma and Welling, 2014; Rezende et al., 2014):

$$\begin{aligned} \mathbf{z}_i &\sim N(\mathbf{0}, \mathbf{I}), \\ \mathbf{x}_i | \mathbf{z}_i &\sim N(\mu_\theta(\mathbf{z}_i), \mathbf{\Sigma}), \quad i = 1, \dots, N, \end{aligned}$$

where the noise variance is $\mathbf{\Sigma} = \text{diag}\{\sigma_j^2\}_{j=1}^G$. The latent variables θ of the neural network are generally estimated via MAP estimation.

An alternative nonlinear mapping is a “skip” or residual neural network $\mu_\theta^{\text{SKIP}} : \mathbb{R}^K \rightarrow \mathbb{R}^G$ (Dieng et al., 2019). It includes direct connections to the latent variables $\mathbf{z}$ in each hidden layer of the neural network. Specifically, the skip neural network μ_θ^{SKIP} has the form:

$$\mu_\theta^{\text{SKIP}}(\mathbf{z}) = W_{L+1}^{(h)} \mathbf{h}_L + W_{L+1}^{(z)} \mathbf{z} \quad (30)$$

$$\text{where } \mathbf{h}_l = a_{b_l}(W_l^{(h)} \mathbf{h}_{l-1} + W_l^{(z)} \mathbf{z}), \quad l = 1, \dots, L, \quad (31)$$

$$\text{with } \mathbf{h}_0 = \mathbf{0}_K. \quad (32)$$

The skip neural network can be used in place of μ_θ in the DGM in Eq. 30. We refer to this model as a skip-DGM.

For both the DGM and skip-DGM, posterior inference is intractable. We fit the DGM using a variational autoencoder (VAE, Kingma and Welling, 2014; Rezende et al., 2014), which optimizes an approximation of the regularized likelihood that uses a variational approximation of the posterior of $p(\mathbf{z}_i | \mathbf{x})$. The variational family is

$$q_\phi(\mathbf{z}_i | \mathbf{x}_i) \sim N(\mu_\phi(\mathbf{x}_i), \sigma_\phi^2(\mathbf{x}_i) \odot \mathbf{I}), \quad i = 1, \dots, N \quad (33)$$

where $\mu_\phi : \mathbb{R}^G \rightarrow \mathbb{R}^K, \sigma_\phi^2 : \mathbb{R}^G \rightarrow \mathbb{R}^K$ are neural networks parameterized by ϕ in a similar way to Eq. 29. The parameters θ and ϕ are estimated by optimizing the evidence lower bound (ELBO) (Kingma and Welling, 2014; Rezende et al., 2014). Note that the parameters θ and ϕ are shared across all N samples and corresponding latent variables unlike mean-field variational Bayes (Jordan et al., 1999; Blei et al., 2017), where each sample has a unique variational parameter. This sharing of functional parameters across

samples is referred to as amortized variational inference (Gershman and Goodman, 2014).

Similarly, we fit the skip-DGM with a skip-VAE, which includes direct connections to the observed data $\mathbf{x}$ in the mappings μ_ϕ^{SKIP} and $\sigma_\phi^{\text{SKIP}}$ (Dieng et al., 2019).

For all neural networks, we use a 3-layer neural network with 20 neurons in each hidden layer. We use a rectified linear unit (ReLU) activation in each of the hidden layers, given by $a_{b_l}(\mathbf{z}) = \max(\mathbf{z} + b_l, 0)$. To estimate the neural network parameters which maximize the ELBO, we use stochastic gradient descent with Adam (Kingma and Ba, 2014) with learning rate 1×10^{-3} .

Synthetic Data

We first consider a linear simulation setting where we would expect PPCA to find an appropriate mapping from the latent space to the observed data, and the DGM and skip-DGM to perform similarly well. We set the number of samples to $N = 1000$, the number of observed features to $G = 10$ and the latent dimension as $K = 2$. The data is generated as

$$\mathbf{x}_i = \mathbf{W}\mathbf{z}_i + \boldsymbol{\varepsilon}_i,$$

where $\mathbf{z}_i \sim N(0, \mathbf{I})$, $\boldsymbol{\varepsilon}_i \sim N(0, \sigma^2 \mathbf{I})$ with true $\sigma^2 = 1$. (Note however that σ^2 is treated as unknown in the inference stage). The matrix $\mathbf{W}$ is the following block matrix,

$$\mathbf{W}^\top = \begin{pmatrix} 5 & 5 & 5 & 5 & 5 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 5 & 5 & 5 & 5 & 5 \end{pmatrix}.$$

That is, the first five values of $\mathbf{x}_i$ are linearly related to the first factor, and the last five values of $\mathbf{x}_i$ are linearly related to the second factor. We generate three datasets: $\{\mathbf{x}_{\text{in}}, \mathbf{x}_{\text{out}}, \mathbf{x}_{\text{val}}\}$.

Model Checking

For both the PPN study, we use a heldout diagnostic

$$D_A(\mathbf{x}; \mathbf{x}_{\text{val}}) = d_A(\mathbf{x}, \hat{\boldsymbol{\theta}}_A), \quad \hat{\boldsymbol{\theta}}_A = \max_{\boldsymbol{\theta}} \log p_A(\boldsymbol{\theta} | \mathbf{x}_{\text{val}}). \quad (34)$$

That is, $\hat{\boldsymbol{\theta}}_A$ is the maximum *a posteriori* estimate for model A given the validation data $\mathbf{x}_{\text{val}}$. The underlying realized diagnostic, $d_A(\mathbf{x}, \hat{\boldsymbol{\theta}}_A)$ is the reconstruction loss,

$$d_A(\mathbf{x}, \hat{\boldsymbol{\theta}}_A) = \sum_{i=1}^n \|\mathbf{x}_i - \mathbb{E}[\mathbf{x}_i | \hat{\boldsymbol{\theta}}_A]\|^2. \quad (35)$$

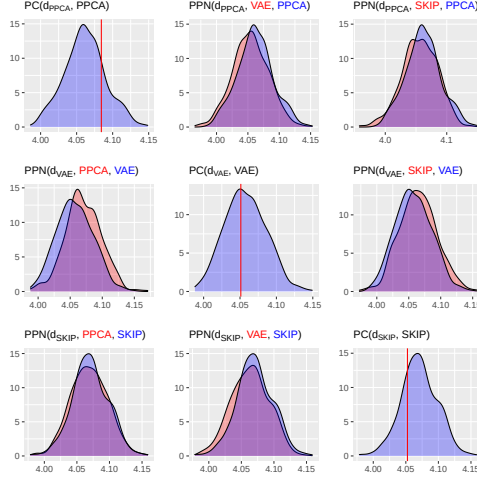


Table 2:: Symmetrized Kullback-Leibler distance between distributions.

	PPCA-2	VAE	SKIP
PPCA-2		0.22	0.27
VAE	0.41		0.25
SKIP	0.22	0.19	

Figure 6: A PPN study of factor analysis models. This study suggests that a linear model is consistent with the data (nonlinear models are not needed). On the diagonal are heldout predictive checks displaying $p_A(d_A(\mathbf{x}_{\text{rep}}^A; \mathbf{x}_{\text{val}})|\mathbf{x}_{\text{in}})$ (blue histogram) and $d_A(\mathbf{x}_{\text{out}}; \mathbf{x}_{\text{val}})$ (red line). All models pass the checks. Beside the diagonal are PPNs, each one checking if a different model can fool the model under study. Specifically, the PPNs compare $p_B(d_A(\mathbf{x}_{\text{rep}}^B; \mathbf{x}_{\text{val}})|\mathbf{x}_{\text{in}})$ (red histogram) and $p_A(d_A(\mathbf{x}_{\text{rep}}^A; \mathbf{x}_{\text{val}})|\mathbf{x}_{\text{in}})$ (blue histogram). The PPN study shows all models (PPCA, VAE, SKIP-VAE) fool all other models.

After fitting the models, each of PPCA, DGM and skip-DGM pass their partial predictive checks (Figure 6). It is unclear which model to choose. The PPN study can help. Consider Figure 6: from both visual inspection and the symmetrized KL divergences (Table 2), the PPN suggests that each model fools every other model. The PPN study concludes that both PPCA and the deep generative models fit the data adequately and can be considered equivalent based on their posterior predictive distributions. If we prefer an interpretable linear model, then we should choose PPCA.

Note we do not consider the Bayes Factors for model comparisons here as they cannot be computed for the deep generative models.

3.3 Nonlinear Factor Analysis

Now consider a simulation setting where the true mapping from the factors to the observed data is nonlinear. In this situation, we expect that both the DGM and the skip-DGM will reconstruct the data well while PPCA will require a larger number of latent dimensions to model the nonlinear mapping.

We set the number of samples to $N = 1000$, the number of observed features to $G = 7$ and the latent dimension to $K = 2$. The data is generated from

$$\mathbf{x}_i = (7z_{i1}, 6z_{i1}, 5z_{i1}^2, 4z_{i2}, 3z_{i2}, 2\sin(\pi/2 \cdot z_{i2}), 1z_{i1} \cdot z_{i2})^\top + \boldsymbol{\epsilon}_i, \quad (36)$$

where $\mathbf{z}_i \sim N(0, \mathbf{I})$, $\boldsymbol{\epsilon}_i \sim N(0, \sigma^2 \mathbf{I})$ with true $\sigma^2 = 1$. That is, the first three columns